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Submission type: Original Investigation

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ABSTRACT

Purpose: American football is widely played by collegiate student-athletes throughout the United States; however, the associated injury risk is greater than in other team sports. Numerous factors likely contribute to this risk yet research identifying these risk factors is limited. The present study sought to explore the relationship between playing experience and position on injury risk within NCAA Division I collegiate footballers. **Methods:** Seventy-six male collegiate student athletes within the American football program of an NCAA Division I university participated. Injuries were recorded over two consecutive seasons. Players were characterized based on college year (freshman, sophomore, junior or senior) and playing position. The effect of playing experience and position on injury incidence rates was analysed using a generalized linear mixed-effects model, with a Poisson distribution, log-linear link function, and offset for hours of training exposure or number of in-game plays (for training and game injuries, respectively). **Results:** The overall rates of non-time loss and time loss game related injuries were 2.1 (90% CI: 1.8-2.5) and 0.6 (90% CI: 0.4-0.8) per 1000 plays respectively. The overall rates of non-time loss and time loss training related injuries were 26.0 (90% CI: 22.6-29.9) and 7.1 (90% CI: 5.9-8.5) per 1000 hours respectively. During training, seniors and running backs displayed the greatest risk. During games, sophomores, juniors and wide receivers were at greatest risk. **Conclusions:** Being aware of the elevated injury risk experienced by certain player groups may help coaches make considered decisions related to training design and player selection.

Key words: American football, gridiron, college, injuries, risk factors

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INTRODUCTION

The potential for physical injury is an accepted risk that differs in size across all sports. The cost of sustaining an injury is multifaceted and the burden is shared among numerous parties, not least the athlete themselves. Consequences incorporate financial,[1] long-term health,[2] emotional and entertainment value elements. American football is widely played by high school and college athletes throughout the United States.[3] However, due to the inherently aggressive and intense physical demands of the game, injuries are a well-acknowledged aspect of the sport. For instance, a total of 394,350 emergency department visits exclusively related to American football participation, among athletes aged 19 and below, were recorded in the United States in 2012.[3] The associated financial burden on healthcare providers related to American football is clearly substantial. American universities also invest significant sums of money into their collegiate athletic programs in an effort to attract and support the best student-athletes. Previous studies have shown that sports teams who incur fewer injuries are more likely to win.[4] Thus, given the financial savings,[5] responsibility to protect student athletes' health and welfare[6] and likelihood of improved performance outcomes,[4] it would seem beneficial to investigate practical solutions to reduce the number and cost of time-loss injuries with relevance to college football.

College football poses a unique sporting environment. Players must combine academic study with a training and playing schedule similar to professional sport. There is also a strict window of eligibility limited to five years. The constraints related to player eligibility create a conundrum for coaches: how to structure training that caters for both relatively novice athletes *and* those that have been within the program for several years and possess significant training/playing experience. Fortington and colleagues[7] observed a greater risk of injury among emerging Australian Rules football players compared to those with three or more years professional playing experience highlighting the potential influence of this

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factor on injury risk. In addition to this challenge, American football is characterized by disparate playing positions and athlete somatotypes,[8] further complicating the issue of training program design.[9] Unsurprisingly, playing position may be influential regarding injury risk.[10] Numerous other injury risk factors have been proposed and investigated within American football including: high volume and low variation of inertial load,[11] poor movement quality[12] and previous injury.[13] Clearly, the causes of sports injury are multifaceted.[14] However, a paucity of research investigating the relationship between playing experience, position and injury exists. Both factors are easily identifiable characteristics that may be worth considering when designing training programs concerned with limiting injury risk.

The aim of the present study was to investigate the influence of playing experience on injury likelihood within a National Collegiate Athletic Association (NCAA) Division I college football program. Additionally, the influence of playing position on injury likelihood was explored. We hypothesized that both playing experience and position would be influential with less experienced players at greater risk of injury. A novel aspect of the present study relates to the statistical analysis performed. Multiple injuries to the same player are often not accounted for in such prospective studies of team ball sport players.[15] However, our analysis not only factored in such events but also accounted for both the training and game exposure of each player.

METHODS

Subjects

Seventy-six male collegiate student athletes within the American football program from an NCAA division I university participated in the present study (age: 20.2 ± 1.5 y, mass: 101.8 ± 18.7 kg, height: 187.0 ± 8.2 cm). Some individuals (n = 25) were included

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twice (once for each competitive season) resulting in a total of 101 player observations. The cohort included freshman (first year; n=23), sophomore (second year; n=31), junior (third year; n=22) and senior (fourth year; n=25) players; indicating their year of eligibility. The practice of 'red shirting' is often adopted within collegiate programs and refers to delaying the playing eligibility window for an individual while still allowing them to train within the program. 'Redshirt' players were characterized to years of experience accordingly (for example, a 'redshirt freshman' who had spent the previous playing season as a 'redshirt' was classified as a sophomore since they were, in fact, in their second year within the program). Prior to enrolling at the University, all players signed an informed consent form indicating that de-identified performance, wellness and injury data collected as part of their athletic participation may be used for research purposes. All experimental procedures were approved by the University Research Compliance Services and conformed to the declaration of Helsinki.

Design

A prospective cohort design was adopted for the present study, which included players whom were on the playing roster for two consecutive seasons. Injury surveillance was performed over the entirety of both seasons with all injuries diagnosed and recorded by certified university athletic trainers and licensed medical staff. Players were characterized based on college year (freshman, sophomore, junior or senior) and position (defensive lineman, DL; defensive back, DB; linebacker, LB; offensive lineman, OL; quarterback, QB; running back, RB; tight end, TE; wide receiver, WR) when assessing injury risk. Specialist players (kickers, punters, long snappers) were not included in the analysis. Where a player eventuated playing two different playing positions, the dominant position (in terms of playing time) was listed. This was only the case for two players. Both training minutes and game

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involvement (number of snaps (or 'plays') participated in per game expressed as a total of offensive, defensive and special team snaps) were accounted for to ensure injury incidence was not simply a feature of greater risk exposure among certain groups.

Methodology

On-field training exposure was recorded in minutes for each player by sport science staff present at every session. The data analyzed consisted of 33 weeks including; all practice sessions during two consecutive seasons' three-week pre-season ('Fall Camp'); and the three main weekly practice sessions and game day during the 14-week (season 1) and 13-week (season 2) in-seasons. In accordance with the team's sport science monitoring policy, data were not collected during the biweekly walk-through/low duration sessions or rest days. In addition, the number of snaps each player took part in during games was recorded and used as the measure of game exposure. For the purposes of the present study an injury was defined as any physical complaint reported to athletic training staff by a player regardless of whether it resulted in time-loss or not (missed training or games). Injuries were categorized as training- or game-related and within each category further delineated into time loss (at least one missed training session or game due to the injury) and non-time loss. Both contact and non-contact injuries were included within the present study. All injury reports were collated and organized in an electronic database. All injury diagnoses made by athletic training staff were reviewed retrospectively and confirmed or amended by a sports physician.

Statistical Analysis

All estimations were made using the *lme4* package[16] with R (version 3.3.1, R Foundation for Statistical Computing, Vienna, Austria). The effect of playing experience and position on injury incidence rates was analysed using a generalized linear mixed-effects model (GLMM), with a Poisson distribution, log-linear link function, and offset for hours of

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training exposure or number of game snaps (for training and game injuries, respectively). This mixed effects model was selected for its ability to account for repeated measurements, and to explore individual differences in injury risk. Rate ratios (RR) and 90% confidence limits (90% CL) for injury incidence rates were generated for each level of playing experience and positional group, with the freshmen and DL groups serving as reference categories.

Inferences regarding the effects of playing experience and position (verses the reference categories) were assessed against a pre-defined smallest worthwhile effect in injury outcome, using a spreadsheet for deriving a confidence interval and clinical inference from a *P*-value.[17] The smallest worthwhile beneficial effect was given by an RR of 0.90 (i.e., a 10% lower injury rate), and conversely the smallest worthwhile harmful effect was given as an RR of 1.11 (i.e., an 11% increase), as previously established.[18] Effects were classified as clear if the percentage likelihood that the true effect was beneficial (i.e., RR below 0.90) was greater than 25%, and the odds ratio between benefit and harm was greater than 66, otherwise the effect was deemed unclear. Effects were qualified against pre-defined probabilistic terms from the following scale: <0.5%, most unlikely; 0.5-5%, very unlikely; 5-25%, unlikely; 25-75%, possibly; 75-95%, likely; 95-99.5%, very likely; >99.5%, most likely.[19] Such analysis is not only unique within the scientific research regarding American football but may also offer practical solutions for assisting coaches and medical staff create and deliver training programs that mitigate injury risk.

RESULTS

The overall rates of non-time loss and time loss game related injuries were 2.1 (90% CI: 1.8-2.5) and 0.6 (90% CI: 0.4-0.8) per 1000 snaps respectively. The relative risks

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associated with each playing experience category and playing position for game related non-time loss and time loss injuries are presented in Tables 3 and 4 respectively.

The overall rates of non-time loss and time loss training related injuries were 26.0 (90% CI: 22.6-29.9) and 7.1 (90% CI: 5.9-8.5) per 1000 hours respectively. The relative risks associated with each playing experience category and playing position for training related non-time loss and time loss injuries are presented in Tables 5 and 6 respectively.

DISCUSSION

The influence of playing experience and position on injury among NCAA Division I collegiate American football players was explored in the present study. The incidence of injury within our American football cohort was high (7.1. injuries per 1000 training hours), especially when comparing to other football codes (e.g. rugby union; 3.0 injuries per 1000 training hours).[20] Both playing experience and position influenced injury risk; however, the strength of the relationships varied between years of experience, positions and whether training or games were considered. Sophomores and juniors were likely at greater risk of time-loss injury during games than freshmen (Table 4). In contrast, seniors were likely at greater risk of injury during training when compared to freshmen (Tables 5 & 6). When considering playing position, TE was broadly the position at lowest risk with RB and WR displaying elevated risk depending on whether training or game contexts were considered (Tables 3, 5 & 6). Comparison of the overall injury rates observed in the present study to those previously reported within American football literature is challenging due to the use of differing reporting metrics. For example, injury incidence within American football research has typically been reported per 1000 athlete exposures rather than per 1000 hours of participation as is more commonly used within other sports.[21] We chose to report injury incidence in training per 1000 hours and per 1000 snaps for games. Whilst it is acknowledged

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that this represents a shift from typical reporting practice within American football research it should also be recognized as a progressive step forward.[22] Using number of 'athlete exposures' is clearly easier from an administrative point of view than quantifying actual training/playing time; however, it raises the question: what is the duration of 'an exposure'? Indeed, Kerr[22] demonstrated how using actual exposure time provides a more accurate estimation of injury risk than simply using 'athlete exposures'. The injury incidence metrics presented herein provide further clarity and detail compared to similar data that have been previously published.

Playing experience

In relation to non-time loss game related injuries playing experience did not have an influence (Table 3). In contrast, sophomore and junior players were likely at an increased risk of sustaining a time loss injury compared to freshmen (Table 4). Why this trend did not extend to senior players remains unclear. At first glance, this could be interpreted as a consequence of the greater game exposure experienced by sophomores and juniors compared to freshman. However, game exposure was accounted for within the statistical analysis precluding this explanation. Indeed, if this were a contributing factor then one would expect seniors to also be at greater risk compared with freshmen. The reason why this risk trend did not extend to senior players may be explained by the increases in physical strength and improvement in technique seen over the course of a player's career.[23] Well-developed fitness has been reported as protective against injury within rugby league.[24, 25] Furthermore, higher levels of strength and fitness have been shown to mitigate the physical fatigue induced by intense exercise and this may potentially manifest as superior resilience to injury.[26] It should be noted that the physical condition of each player was not included within our analyses and as such its influence on injury risk within the present cohort was unclear. More research

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investigating the potential protective influence of highly developed physical qualities in relation to injury is warranted. Furthermore, given the challenge of adapting to a new game plan, coach and training program in their first year of eligibility it is possible that only the very best, exceptional freshmen play in games. One aspect that makes these freshmen exceptional may be extremely well developed physical qualities and outstanding technical skill, both attributes that may help protect them against injury.[24, 27]

When considering both non-time loss and time loss training injuries playing experience did appear to have an influence, with seniors likely at increased risk compared to freshmen (Tables 5 & 6). This finding contradicts our hypothesis that less experienced players would be at greater risk of injury. As aforementioned, new players entering a collegiate football program are confronted with arduous physical training regimens that represent a step up from the high school environments from which they are recruited each year. Therefore it is reasonable to expect that lack of adequate physical adaptation to the increased training/game demands may have predisposed freshmen to injury more so compared to seniors. This was indeed the case among Australian Rules football players[7]; however a similar trend was not observed in soccer.[28] The practice of 'red shirting' is often adopted within collegiate programs and refers to delaying the playing eligibility window for an individual while still allowing them to train within the program. The intention behind red shirting is to provide a player with a year to train with the team but not play in games; in an effort to aid the transition between high school and collegiate football. One facet of this transition involves the physical conditioning and adaptation of the player. The present results provide an indication that this practice may be effective for managing injury risk. Equally, the academic workload experienced by seniors will most likely be greater than freshmen and related stress has been reported as a contributor to injury risk.[29] A further explanation may relate to the specific content of training sessions and potential differences between freshmen

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and seniors. Seniors may be more likely to participate in full-contact drills where the emphasis is focused upon the development of physical qualities and executing tactical directives at, or close to, game intensity. In contrast, freshmen may generally spend a larger percentage of their sessions learning tactical plays and receiving extensive instruction from coaches. However, training content was not empirically analyzed in the present study. Indeed, this latter theory is somewhat speculative and further research quantifying the typical loads experienced by different player groups based on experience and position during training would be extremely insightful.

Playing position

The vastly different sizes, body composition profiles and physical demands apparent amongst the numerous disparate playing positions mean it is conceivable that injury risk profiles may vary between them.[8] When considering non-time loss game-related injuries, DB and TE players likely suffered fewer injuries than DL players (Table 3). In contrast, WR players displayed the highest injury risk of all positions. WR players may display the greatest risk for non-time loss injury due to the nature of the tackles they often sustain (for example, moderate impact collisions).[9] This may be further exacerbated by the fact WR players typically display the lowest absolute strength scores and body fat percentages of all playing positions.[8] Thus WRs may be more vulnerable to the detrimental physical effects of game-related fatigue and potentially injury than other positions.[26] However, the frequency and intensity of the collisions experienced by the WRs in the present cohort were not quantified. Similarly, the physical qualities of the WRs were not considered within our statistical analyses, limiting this explanation as speculative. Furthermore, Wellman et al.[30] reported that WRs covered the greatest total, high intensity running and sprint distance of all offensive positions. Gabbett and Ullah[31] suggested that sprint distance may be related to soft-tissue

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injury; hence, this could be another contributor to the elevated risk observed among WRs. However, the sprinting demands of the WR position may be of little concern if appropriate training has been completed by the player whereby any in-game sprinting does not represent a 'spike' in sprinting load.[32]

Interestingly, no influence of playing position was observed for time loss injuries sustained during games (Table 4). In the case of some positions (TE and QB), too few injuries were observed to allow statistical inferences to be calculated. The relatively low number of games (27) included within the observation period may have contributed to the unclear findings. Future studies conducted over longer observation periods or in collaboration between multiple football programs may reveal further insights related to the relationship between playing position and game related injury risk.

Regarding time loss injuries sustained during training, QB and TE players were likely at lower risk than DL players (Table 6). QBs arguably represent the most important player in any offense since they are directly involved in every play. Thus, in an effort to protect them from collision injuries, QBs commonly wear different coloured jerseys during training, which are aimed at directing opposing teammates to avoid contact. In comparison, the TE position arguably represents one of the most physically and technically well rounded positions with these individuals requiring the capacity to act as an auxiliary OL or WR on occasion depending on the play call. As a result, it is possible that the types of individuals who play the TE position display particularly well developed athletic qualities (whether prior to enrollment or developed during their collegiate career), such as strength and size, that allow them to fulfill multiple roles within the team, which may contribute to injury resilience. However, this assertion is speculative and further research is required to substantiate such a theory.

Of all playing positions RBs were at the greatest risk of time loss injury during training (Table 6). The RB position inherently involves some of the most significant physical

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collisions experienced by any player.[9] This may explain why they likely suffer more training time loss injuries than any other position. The frequency and intensity of collisions experienced by the participants observed in the present study were not quantified. As such, it cannot be empirically confirmed whether differences in collisions between position groups contributed to injury incidence in the present study. Indeed, the demands of the RB position[9] provide coaches with a cost vs benefit conundrum: how best to prepare the players for the substantial collisions they will inevitably experience during games while reducing the likelihood of suffering a preventable contact injury during training. Strength training during the off-season may represent a low-risk strategy to improve physical qualities, such as increasing lean muscle mass, likely to offer protection against injury during collisions. More research which evaluates the differences in contact and collision forces in training vs games, whilst also identifying the mechanisms behind related injuries may help to optimize preparation for these challenging scenarios.

Limitations

A number of limitations should be acknowledged when interpreting the present results. Only one collegiate program was included, with the observation period encompassing only two seasons. As a result, it is somewhat unclear how generalizable the conclusions made herein may apply to other programs due to differing coaching styles and game preparation philosophies. The relatively small sample size and resultant number of injuries recorded may explain the frequency of 'unclear' inferences reported in Tables 3, 4, 5 and 6. Furthermore, while we distinguished between time loss and non-time loss injuries in addition to categorizing based on whether they were sustained during training or games we did not create contact/non-contact, soft tissue or discrete severity categories which limits some of conclusions and assumptions. However, this would have resulted in too low a frequency of

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injury events to make any meaningful statistical inferences. Collaborative research between institutions incorporating data sharing and conducted over longer observation periods would provide further insights into the relationship between playing experience, position and injury.

PRACTICAL APPLICATIONS

Seniors appear to be at greatest risk of injury during training. Coaches may consider adapting sessions to limit the exposure of seniors to the most injurious elements of training, for example: contact drills. Similarly, athletic training staff may wish to pay particular attention to targeted recovery strategies for senior players or strength and conditioning staff may look to enhance the physical attributes of these players to protect against injury. RBs displayed the highest risk of time loss injury during training. Due to the game demands of the RB position substantial collisions are inevitable[9]; however, being aware of the elevated injury risk experienced by these players during training may help coaches make considered decisions when designing sessions. It is worth noting that reducing the risk of injury during training needs to be balanced against the need to condition players for the game demands.

CONCLUSIONS

Both experience and position influence injury risk during games and training. Training should be designed to try and mitigate these influences whilst balancing the need to adequately prepare and condition the players for the holistic demands of the game. Reporting injury incidences per 1000 hours and per 1000 snaps, as we have herein, provides greater detail than reporting per 1000 athlete exposures, as is currently common practice within American football research. Furthermore, the use of magnitude-based inferences represents a progressive step within American football injury research. Future studies may seek to replicate the present investigation across multiple collegiate programs and over longer observation periods.

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"Playing Experience and Position Influence Injury Risk Among NCAA Division I Collegiate Footballers"
by McCunn R et al.
International Journal of Sports Physiology and Performance
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Table 1: Descriptive injury data based on playing experience

Playing experience	Sum of training minutes	Sum of snap counts	Sum of all time loss training injuries	Sum of all time loss game injuries	Sum of all non-time loss training injuries	Sum of all non-time loss game injuries	Sum of all non-contact time-loss injuries
2014							
Freshmen	115413	5109	8	1	45	13	6
Sophomore	84351	3212	9	2	39	11	6
Junior	97471	7576	10	6	50	21	7
Senior	104974	8717	17	4	59	15	11
2015							
Freshmen	46244	1060	7	0	18	0	3
Sophomore	137485	8778	20	6	52	18	10
Junior	77672	5723	8	4	30	12	8
Senior	82181	7669	9	4	36	12	6

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Table 2: Descriptive injury data based on playing position

Playing position			Sum of training minutes	Sum of snap counts	Sum of all time loss training injuries	Sum of all time loss game injuries	Sum of all non-time loss training injuries	Sum of all non-time loss game injuries	Sum of all non-contact time-loss injuries
Defense			414274	27755	55	17	199	49	37
	DL								
		2014	90520	5934	10	6	46	16	8
		2015	59451	3747	9	1	26	4	3
	DB								
		2014	64470	4153	9	1	39	7	6
		2015	62245	5027	10	1	25	3	6
	LB								
		2014	76985	4427	11	4	40	10	9
		2015	60603	4467	6	4	23	9	5
Offense			331517	20089	33	10	130	53	20
	OL								
		2014	46921	4046	5	0	21	6	1
		2015	46541	4200	3	5	15	13	3
	QB								
		2014	16661	1105	0	0	3	3	0
		2015	20348	1128	1	0	5	3	0

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Table 3: Non-time loss injury risk factors during games

Predictor variable		Relative risk	Lower CL (90%)	Upper CL (90%)	Inference	P-value
Playing position						
	DL (n = 19)	1.00 (ref)	-	-	-	-
	DB (n = 18)	0.54	0.28	1.06	Likely ↓	0.13
	LB (n = 19)	1.06	0.60	1.85	Unclear	0.88
	OL (n = 13)	1.10	0.62	1.94	Unclear	0.78
	QB (n = 5)	1.22	0.54	2.78	Unclear	0.69
	RB (n = 8)	1.38	0.65	2.93	Unclear	0.48
	TE (n = 5)	0.24	0.05	1.43	Likely ↓	0.19
	WR (n = 14)	1.84	1.01	3.36	Likely ↑	0.10
Playing experience						
	Freshmen (n = 23)	1.00 (ref)	-	-	-	-
	Sophomore (n = 31)	1.39	0.78	2.46	Unclear	0.35
	Junior (n = 22)	1.45	0.79	2.64	Unclear	0.31
	Senior (n = 25)	0.99	0.53	1.84	Unclear	0.99

CL; confidence limit, DB; defensive back, DL; defensive lineman, LB; linebacker, OL; offensive lineman, QB; quarterback, RB; running back, TE; tight end, WR; wide receiver

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Table 4: time loss injury risk factors during games.

Predictor variable		Relative risk	Lower CL	Upper CL	Inference	P-value
Playing position						
	DL (n = 19)	1.00 (ref)	-	-	-	-
	DB (n = 18)	0.33	0.09	1.24	Unclear	0.17
	LB (n = 19)	1.22	0.52	2.85	Unclear	0.71
	OL (n = 13)	0.86	0.33	2.28	Unclear	0.80
	*QB (n = 5)	-	-	-	Unclear	-
	RB (n = 8)	1.04	0.27	4.04	Unclear	0.96
	*TE (n = 5)	-	-	-	Unclear	-
	WR (n = 14)	0.95	0.29	3.08	Unclear	0.95
Playing experience						
	Freshmen (n = 23)	1.00 (ref)	-	-	-	-
	Sophomore (n = 31)	4.43	0.77	25.49	Likely ↑	0.16
	Junior (n = 22)	5.08	0.88	29.41	Likely ↑	0.13
	Senior (n = 25)	3.00	0.50	17.97	Unclear	0.31

* Risk could not be calculated for QB and TE positions due to low injury counts.

CL; confidence limit, DB; defensive back, DL; defensive lineman, LB; linebacker, OL; offensive lineman, QB; quarterback, RB; running back, TE; tight end, WR; wide receive

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Table 5: Non-time loss injury risk factors during training.

Predictor variable	Relative risk	Lower CL	Upper CL	Inference	P-value
Playing position					
DL (n = 19)	1.00 (ref)	-	-	-	-
DB (n = 18)	1.18	0.78	1.81	Unclear	0.51
LB (n = 19)	0.99	0.64	1.51	Unclear	0.96
OL (n = 13)	0.85	0.52	1.38	Unclear	0.58
QB (n = 5)	0.43	0.20	0.93	Likely ↓	0.07
RB (n = 8)	1.16	0.68	1.97	Unclear	0.65
TE (n = 5)	0.50	0.24	1.03	Likely ↓	0.12
WR (n = 14)	1.02	0.65	1.62	Unclear	0.94
Playing experience					
Freshmen (n = 23)	1.00 (ref)	-	-	-	-
Sophomore (n = 31)	1.04	0.76	1.42	Unclear	0.84
Junior (n = 22)	1.27	0.88	1.83	Unclear	0.28
Senior (n = 25)	1.37	0.96	1.96	Likely ↑	0.14

CL; confidence limit, DB; defensive back, DL; defensive lineman, LB; linebacker, OL; offensive lineman, QB; quarterback, RB; running back, TE; tight end, WR; wide receiver

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Table 6: Time loss injury risk factors during training.

Predictor variable	Relative risk	Lower CL	Upper CL	Inference	P-value
Playing position					
DL (n = 19)	1.00 (ref)	-	-	-	-
DB (n = 18)	1.31	0.76	2.27	Unclear	0.42
LB (n = 19)	0.96	0.55	1.66	Unclear	0.90
OL (n = 13)	0.71	0.35	1.41	Unclear	0.41
QB (n = 5)	0.21	0.04	1.12	Likely ↓	0.13
RB (n = 8)	1.62	0.86	3.07	Likely ↑	0.21
TE (n = 5)	0.19	0.03	1.01	Likely ↓	0.10
WR (n = 14)	1.02	0.55	1.88	Unclear	0.96
Playing experience					
Freshmen (n = 23)	1.00 (ref)	-	-	-	-
Sophomore (n = 31)	1.55	0.91	2.64	Likely ↑	0.18
Junior (n = 22)	1.53	0.84	2.78	Unclear	0.25
Senior (n = 25)	1.79	1.04	3.10	Likely ↑	0.08

CL; confidence limit, DB; defensive back, DL; defensive lineman, LB; linebacker, OL; offensive lineman, QB; quarterback, RB; running back, TE; tight end, WR; wide receiver

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